ALG-03 Teacher Diagnostic Key

Diagnostic routes, hint ladder and verified answers

RECONNECT → DISCOVER → MAKE SENSE → TRY → DIAGNOSE → FADE → ADOPT → TRANSFER

ALG-03 - Teacher Diagnostic Key

Recognition lab

1. **alpha+beta=7**, **alpha beta=10**; reconstruct target.
2. Factor **(x-5)(x-2)**.
3. **Delta=16-20<0**.
4. Route to ALG-02; complete square **(x-2)^2+1**.
5. **x-3**.
6. Evaluate **P(2)**.
7. **P(x-4)**.
8. Each root moves down by 4.
9. Use **x^2≡1**; remainder **1**.
10. **x^3≡1**; 2026 mod 3 =1, remainder **x**.
11. Subtract/combine to eliminate leading terms.
12. **1/r+1/s=(r+s)/(rs)**; need sum/product.
13. Evaluate at **x=-1**; result 0, so yes.
14. **Delta=0**.
15. For roots doubled, test a scalar-normalized version of **P(x/2)**.
16. No; a divisor/relation is needed to define reduction.

Practice answers

1. **(x-2)(x-3)**, roots 2,3.
2. sum 7, product 10.
3. **29**.
4. **Delta=0**; repeated root 3.
5. **P(1)=8**.
6. sum **5/2**, product **-3/2**.
7. **8^3-3128=224**.
8. **Delta=-4**; no real roots.
9. **Delta=1**; two distinct real roots.
10. **P(-1)=0**; yes.
11. **P(x-4)=(x-6)(x-7)=x^2-13x+42**.
12. roots **-2,-1**.
13. **1**.
14. **x**.
15. **x^5≡55x-21 (mod x^2-3x+1)**.
16. subtract -> **-x+3=0**; common root 3.
17. **(alpha+beta)/(alpha beta)=(-2/3)/(-5/3)=2/5**.
18. **p^3+q^3=s^3-3st**.
19. **Delta=k^2-36=0**; **k=+-6**, repeated roots **+-3** respectively.
20. **P(1)=0**; remainder 0, factor yes.
21. **41=81-2alpha beta**, so product 20; polynomial **x^2-9x+20**.
22. **x^3≡1**; exponents 1000,999,998 reduce to 1,0,2 mod3, so **x+1+x^2≡0**.
23. First at -1: **1-a+6=0** -> **a=7**; second **1-b+3=0** -> **b=4**. Polynomials roots **-1,-6** and **-1,-3**.
24. **20 mod3=2**; reduce **x^2 ≡ -x-1**, so the canonical remainder is **-x-1**.

25. roots **(9-1)/2=4**, **(9+1)/2=5**; polynomial **x^2-9x+20**.
26. Vieta avoids computing roots and preserves only requested symmetric information.
27. root count -> ALG-03 discriminant, no real roots; minimum -> ALG-02 square completion, min 1.
28. Desired roots **alpha-3,beta-3** come from **P(x+3)**.
29. A characteristic relation identifies powers modulo a low-degree polynomial; reduce rather than iterate huge powers.
30. Subtraction/remainder elimination toward a common factor is the elementary form of gcd reasoning.

H0 answers

1. sum **9/2**, product **2**; square sum **81/4-4=65/4**.
2. **Delta=4-60=-56**; no real roots.
3. **Delta=0**; root **12/(8)=3/2**.
4. Original roots 2,4; shifted roots 5,7; polynomial **x^2-12x+35** = **P(x-3)**.
5. Roots move up by 2: if **P(alpha)=0**, **Q(alpha+2)=P(alpha)=0**.
6. **3^4+6+7=94**.
7. **P(2)=8-12+4=0**; yes.
8. **99 mod3=0**; remainder **1**.
9. Relation **x^2=2x+1**; reductions give **x^6=70x+29**.
10. subtract -> **-x+2=0**; common root **2**.
11. sum 10, product 21; **1000-630=370**.
12. ALG-02; **(x-5)^2-4**, minimum **-4**.
13. **Delta=k^2-16>0**, so **|k|>4**.
14. A low-degree relation already identifies the remainder class; direct high-power expansion computes irrelevant information.
15. If old root is **alpha**, **P((alpha+5)+5)=P(alpha+10)** need not be zero; **P(x-5)** is correct for root shift +5.
16. Difference gives a necessary candidate for any common root; original checks establish sufficiency.

Diagnostic tags

- **ALG03-R1** solves roots unnecessarily instead of using invariants.
- **ALG03-R2** discriminant used for an optimization request or vice versa.
- **ALG03-R3** root/input shift sign reversed.
- **ALG03-R4** high powers calculated rather than reduced.
- **ALG03-R5** factor/remainder theorem not connected through evaluation.
- **ALG03-R6** common-root candidate not checked in both originals.